Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 10.1 · Converting and measuring

We use the unit circle to measure angles in radians.

1. Convert $150°$ and $-90°$ to radians, showing the conversion factor.
2. A circular sector has radius 4 cm and an angle of $\frac{\pi}{3}$ radians. Calculate the exact length of its arc.
3. Explain why an angle measure can be negative while a geometric arc length is positive.

Exercise 10.2 · Reducing angles

Consider the real numbers $\frac{17\pi}{6}$, $-\frac{7\pi}{4}$ and 5π.

1. For each, determine a representative in $[0, 2\pi)$ having the same image on the unit circle. Show the number of turns added or removed.
2. Place the three images on a circle and identify them by the starting real numbers.

Exercise 10.3 · Exact values

We look for the exact values of the cosine and sine of $\frac{2\pi}{3}$, $-\frac{\pi}{6}$ and $\frac{5\pi}{4}$.

1. Place the three images on the unit circle. For each, state the expected sign of its two coordinates.
2. Calculate the exact values using a standard angle and a symmetry. Present the answer in a table: angle, cosine, sine.

Exercise 10.4 · Recovering the sign

A real number x satisfies $\sin x = \frac{5}{13}$. Its image on the unit circle lies in the second quadrant, that is to the left of the y-axis and above the x-axis.

1. Determine the exact value of $\cos x$. Justify both its absolute value and its sign.
2. Explain why the value of $\sin x$ alone would not have been enough to choose the sign of $\cos x$.

Exercise 10.5 · A wheel

A wheel of radius 0.30 m rolls on horizontal ground without slipping. It turns through an angle of $\frac{5\pi}{3}$ radians. We look for the distance travelled by its axle in the direction of motion.

1. Calculate this distance in exact form, explaining the link with the arc length unrolled.
2. Explain why the length of a chord of the circle must not be used. Is the path of a point fixed on the rim of the wheel, seen from the ground, the same as that of the axle?

Exercise 10.6 · Chord and scalar product

On the unit circle with centre O, the point A is the image of 0 and the point B that of $\frac{2\pi}{3}$.

1. Determine the exact coordinates of A and B, then calculate the length AB.
2. Recover this length by applying the Al-Kashi relation to the triangle OAB. State the lengths and the angle used.

Exercise 10.7 · Reading an equation on the circle

We look for the real numbers x in the interval $[0, 2\pi)$ whose image on the unit circle has x-coordinate $\frac{1}{2}$.

1. Translate the condition on the x-coordinate into an equation involving $\cos x$.
2. Draw the corresponding vertical line on a circle, mark its intersections and give the two solutions. Check that they belong to the interval required.

Exercise 10.8 · Delayed review

Consider the angle of measure $\frac{25\pi}{4}$ radians. The answers must be exact, with no decimal approximation.

1. Reduce this angle, showing a whole number of complete turns.
2. Deduce its cosine and sine. Justify the values using the unit circle.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 32 p. 220, questions a and b. Use the exact value supplied and the related angles, without a calculator.

CQFD, exercise 39 p. 220, questions 1, 2, 3 and 5. Do the four calculation and circle-reading questions; justify the answers on your paper.